

2021 HKDSE Physics

Paper 2 — Elective Sections

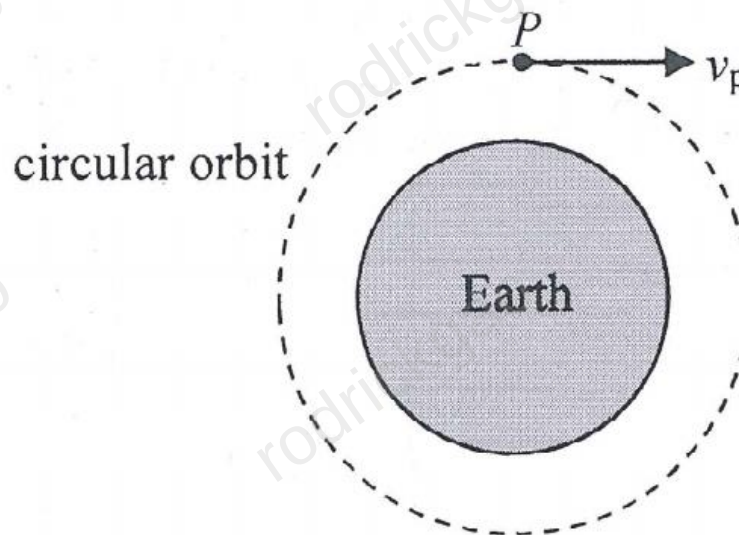
Editable transcription of all questions and formulae. Only necessary, non-duplicated original diagrams are inserted. Each official answer is followed by a verified concise full-mark solution in English, with editable subscripts and superscripts.

Section A: Astronomy

Multiple-choice questions

1.1

An object at P has speed v_p such that $v_1 < v_p < v_2$, where v_1 is the circular-orbit speed through P and v_2 is the escape speed from P. Which statement about its subsequent motion is INCORRECT?



- A It follows an elliptical path.
- B It moves with constant speed.**
- C The farthest point is on the opposite side of the Earth.
- D Its path does not intersect the circular orbit except at P.

Official answer: B

Because $v_1 < v_p < v_2$, the object is gravitationally bound but is not in a circular orbit, so it follows an ellipse and its speed varies around the orbit. The incorrect statement is that it moves with constant speed. Answer: B.

1.2

Planets X and Y orbit the same star. What is $\text{radius}(X)/\text{radius}(Y)$ if $\text{period}(X)/\text{period}(Y) = 8$?

- A $1/4$
- B 4
- C $1/(16\sqrt{2})$
- D $16\sqrt{2}$

Official answer: B

Kepler's third law gives $T^2 \propto r^3$. Therefore $r_X/r_Y = (T_X/T_Y)^{2/3} = 8^{2/3} = 4$. Answer: B.

1.3

Which observations by Galileo are consistent with the heliocentric model but not the geocentric model?

- (1) retrograde motion of Mars
- (2) moons orbiting Jupiter
- (3) the changing phases of Venus

- A (1) only
- B (2) only
- C (1) and (2) only
- D (2) and (3) only

Official answer: D

Moons orbiting Jupiter show that not everything orbits Earth, and the full set of Venusian phases requires Venus to orbit the Sun. Retrograde motion alone can also be modelled geocentrically. Thus (2) and (3) are correct. Answer: D.

1.4

Which statement about apparent magnitude and absolute magnitude is INCORRECT?

- A The absolute magnitude of a star can be larger than its apparent magnitude.
- B The absolute magnitude of a star can be smaller than its apparent magnitude.
- C If the absolute magnitude of one star equals the apparent magnitude of another, the energy received per unit time per unit area from the two stars must be equal.
- D If the apparent magnitude of one star equals that of another, the energy received per unit time per unit area from the two stars must be equal.

Official answer: C

Equal apparent magnitudes mean equal received flux, but equating one star's absolute magnitude with another star's apparent magnitude does not compare them at the same distance. Statement C is therefore incorrect. Answer: C.

1.5

Stars X and Y have equal apparent brightness. The parallax angle of X is twice that of Y. What is $\text{luminosity}(X)/\text{luminosity}(Y)$?

A $1/4$

B $1/2$

C 2

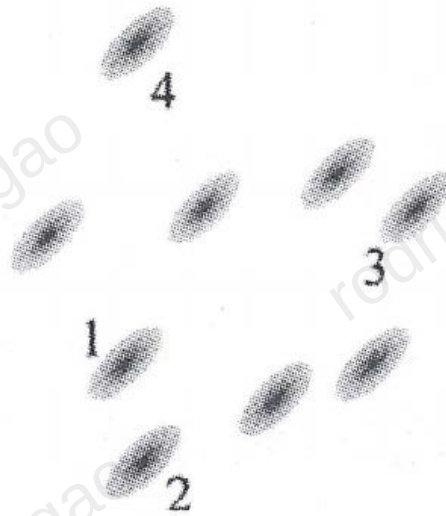
D 4

Official answer: A

Equal apparent brightness gives $L_X/d_X^2 = L_Y/d_Y^2$. Since parallax is inversely proportional to distance, $p_X = 2p_Y$ means $d_X = d_Y/2$; hence $L_X/L_Y = (1/2)^2 = 1/4$. Answer: A.

1.6

Four galaxies are shown in a snapshot. Which statements about their relative motion are correct?



- (1) For observers in galaxy 1, absorption lines from galaxy 4 have a greater redshift than those from galaxy 2.
- (2) For observers in galaxy 2, galaxy 4 is moving away faster than galaxy 1.
- (3) For observers in galaxy 3, galaxies 1 and 4 are moving away at roughly the same speed.

- A (1) only
- B (1) and (2) only
- C (2) and (3) only
- D (1), (2) and (3)

Official answer: D

For uniform cosmic expansion, recession speed is proportional to separation. Applying this from each of the shown galaxies makes statements (1), (2) and (3) consistent with the snapshot. Answer: D.

1.7

What information can be deduced from the absorption spectrum of a star?

- (1) spectral class
- (2) radial velocity
- (3) chemical composition of the core

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

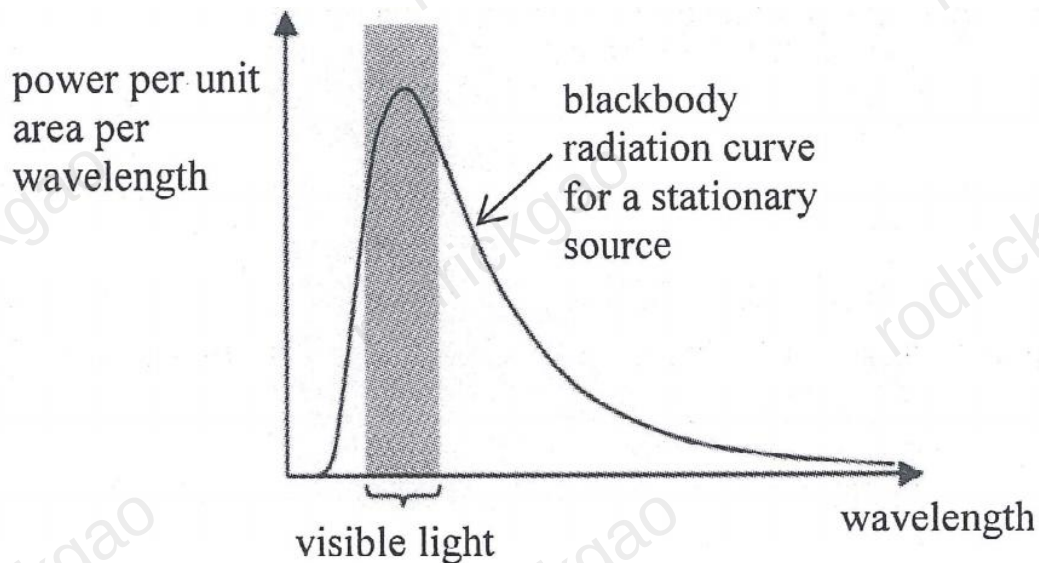
D (1), (2) and (3)

Official answer: A

Absorption-line pattern identifies spectral class, and Doppler displacement gives radial velocity. It reveals atmospheric composition, not the chemical composition of the stellar core. Thus (1) and (2) are correct. Answer: A.

1.8

A blackbody source is moving away from an observer. Which statements about the observed spectrum are correct?



(1) The peak shifts to a longer wavelength.

(2) The source would be inferred to be cooler.

(3) Its observed colour is different.

A (1) only

B (1) and (2) only

C (1) and (3) only

D (1), (2) and (3)

Official answer: D

Recession redshifts the entire blackbody spectrum: the observed peak moves to longer wavelength, Wien's law would imply a lower temperature, and the observed colour changes. All three statements are correct. Answer: D.

Structured question

Sagittarius A* (Sgr A*) lies 7940 pc from Earth. Star X follows an elliptical orbit around it with period 16.0 years.

(a)(i) The semi-major axis subtends 0.125 arc second at Earth. Determine the semi-major axis in AU. (1 mark)

(a)(ii) Using $T^2 = 4\pi^2 a^3 / (GM)$, show that the mass of Sgr A* is 3.82×10^6 solar masses. (2 marks)

- (b) An observer is aligned with the major axis ABCD of the orbit. The graph shows the radial velocity v_r of X, with positive v_r meaning recession. Sgr A* is at position 1 or 2.

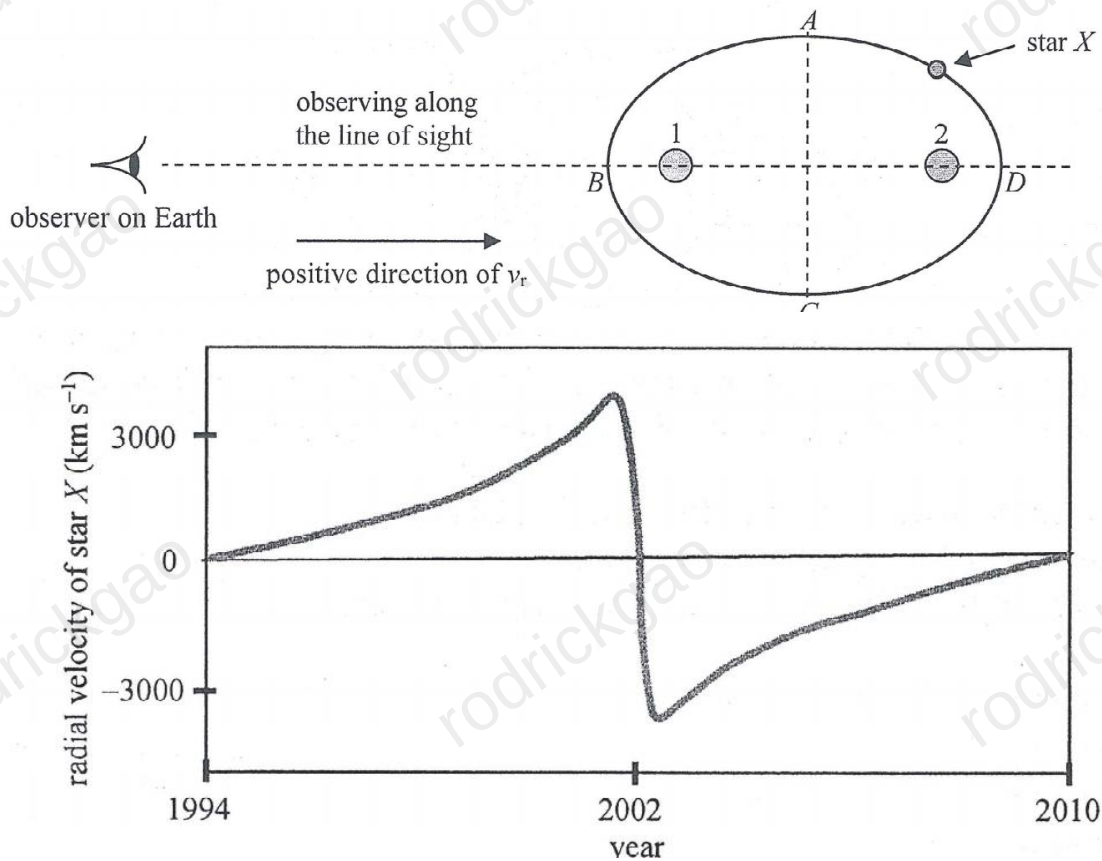


Figure 1.1

- (b)(i) State a method of determining v_r and explain the difference between positive and negative values. (2 marks)
- (b)(ii) State whether X was at A, B, C or D around 2002, and determine whether Sgr A* was at position 1 or 2. Explain. (2 marks)
- (c) The escape speed is $v = \sqrt{2GM/R}$. Since light cannot escape from a black hole, estimate the radius of Sgr A* in AU. Take $GM_{\odot} = 1.33 \times 10^{20} \text{ N m}^2 \text{ kg}^{-1}$. (3 marks)

Official marking scheme

Solution		Marks	Remarks
1. (a) (i)	By definition: $1'' = 1 \text{ AU}/1 \text{ pc}$ Hence, the semi-major axis $a = 0.125'' \times 7940 \text{ pc}$ $= 992.5 \text{ AU} \approx 993 \text{ AU}$	1A	
	Or angular size $\theta = 0.125''$ $= \frac{0.125}{60 \times 60} \times \frac{\pi}{180^\circ}$ $= 6.06 \times 10^{-7} \text{ rad}$ $a = d \times \theta$ $= (7940 \times 206265) \times 6.06 \times 10^{-7}$ $= 992.5 \text{ AU}$	1A	
		1	
(ii)	Consider the Earth and the Sun, we have $a = 1 \text{ AU}$, $M = M_S$ and $T = 1 \text{ year}$. By simple ratio, $\frac{M_{\text{Sgr A}^*}}{M_S} = \frac{\left(\frac{993 \text{ AU}}{1 \text{ AU}}\right)^3}{\left(\frac{16.0 \text{ yr}}{1 \text{ yr}}\right)^2} = 3.819017 \times 10^6 \approx 3.82 \times 10^6$ Hence the mass of the source is about $3.82 \times 10^6 M_S$.	1M 1M	
		2	
(b) (i)	Radial velocity v_r : measured by Doppler shift of spectral lines (emission / absorption lines) of star X . For positive v_r , the observed wavelength of the spectral features will be redshifted (larger wavelength or lower frequency) or blue shift (shorter wavelength or higher frequency) for negative v_r .	1A 1A	
		2	
(ii)	At position D (around 2002). v_r is positive (away from the Earth) and increasing till 2002, which implies that (the massive force centre) Sgr A* is in position 2 (according to Kepler's second law)	1A 1A	Accept: When X is closer to Sgr A*, it moves faster due to very large gravitational force.
	Or In 2002, star X experienced a sharp change in (radial) velocity / very large acceleration, from a large, positive v_r (away from the Earth) to a large, negative v_r (towards the Earth), indicating that it is close to Sgr A* (according to Kepler's second law), Sgr A* is therefore in position 2.	2	
(c)			
As escape velocity $v = \sqrt{\frac{2GM}{R}}$, we have $R = \frac{2GM}{v^2}$. Setting v as the speed of light c , radius of black hole $R_{\text{BH}} = \frac{2GM_{\text{Sgr A}^*}}{c^2}$ $= \frac{2(1.33 \times 10^{20} \times 3.82 \times 10^6)}{(3 \times 10^8)^2}$ $= 1.129022 \times 10^{10} \text{ m} \approx 1.13 \times 10^{10} \text{ m}$ $\approx 0.0752681 \text{ (AU)} \approx 0.075 \text{ (AU)}$		1M 1M 1A	
		3	

(a)(i) At 7940 pc, 0.125 arcsec corresponds to $a = 7940(0.125) = 992.5 \text{ AU} \approx 993 \text{ AU}$.

(ii) Comparing $T^2 = 4\pi^2 a^3 / (GM)$ with Earth gives $M/M_\odot = (992.5^3) / (16.0^2) = 3.82 \times 10^6$.

(b)(i) Measure the Doppler shift of X 's spectral lines; positive v_r gives redshift and recession, while negative v_r gives blueshift and approach.

(ii) Around 2002 X was at D . Its rapid reversal from large positive to large negative v_r shows it passed close to the massive focus, so Sgr A* is at position 2.

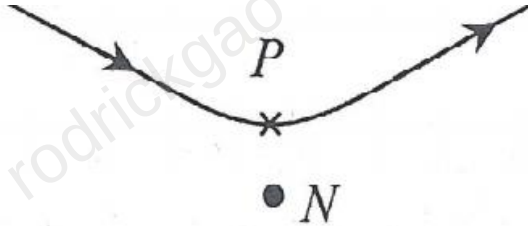
(c) Set escape speed equal to c : $R = 2GM/c^2 = 2(1.33 \times 10^{20})(3.82 \times 10^6) / (3.00 \times 10^8)^2 = 1.13 \times 10^{10} \text{ m} = 0.075 \text{ AU}$.

Section B: Atomic World

Multiple-choice questions

2.1

The path of an α particle approaching a massive nucleus at N is shown. At P the particle is closest to the nucleus. Which statement is correct?



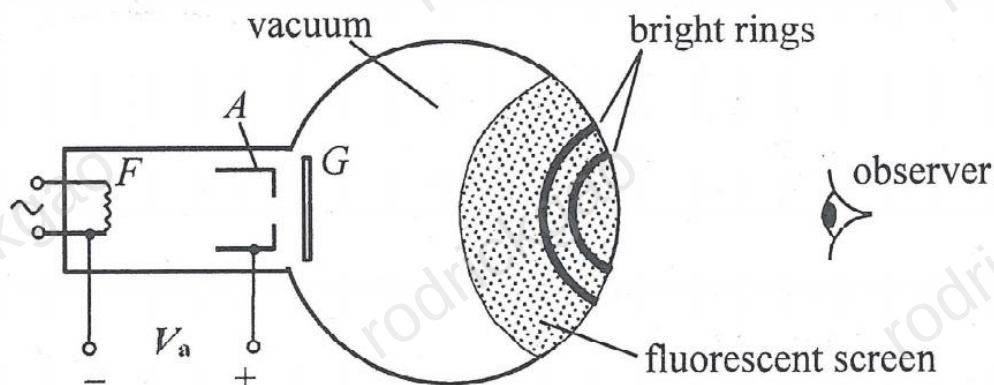
- A At P its kinetic energy is maximum.
- B At P its total energy is minimum.
- C For a nucleus with a larger atomic number, PN would be larger.
- D For a larger initial kinetic energy, PN would be larger.

Official answer: C

A larger nuclear charge gives stronger electrostatic repulsion, so the α -particle turns at a greater closest distance PN. At P the speed and kinetic energy are minimum, while total mechanical energy remains constant. Answer: C.

2.2

An electron diffraction tube produces bright and dark concentric rings. Which descriptions are correct?



(1) It demonstrates the wave nature of fast-moving electrons.

(2) Electrons are diffracted by the graphite film.

(3) If the accelerating voltage V_a increases slightly, the radii of the rings increase.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

Official answer: A

The rings demonstrate the wave nature of fast electrons and arise from diffraction by graphite. Increasing the accelerating voltage reduces the de Broglie wavelength, so the ring radii decrease, not increase. Thus (1) and (2) are correct. Answer: A.

2.3

Monochromatic lights of wavelengths λ and 2λ give maximum photoelectron kinetic energies in the ratio 3: 1.

Find the longest wavelength that can trigger photoemission.

A $5\lambda/2$

B 3λ

C $7\lambda/2$

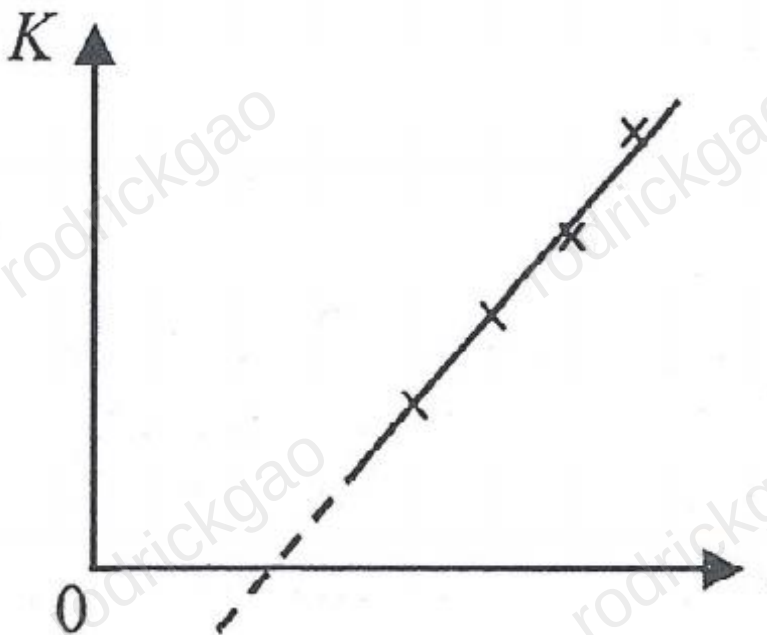
D 4λ

Official answer: D

Let $E = hc/\lambda$. Then $E - \phi = 3[E/2 - \phi]$, giving $\phi = E/4$. The threshold wavelength is $\lambda_0 = hc/\phi = 4\lambda$. Answer: D.

2.4

A graph of maximum photoelectron kinetic energy K against $1/\lambda$ is obtained. Light of lower intensity is then used on another metal of smaller work function. How do the slope and horizontal intercept change?



... if incident light of lower intensity is

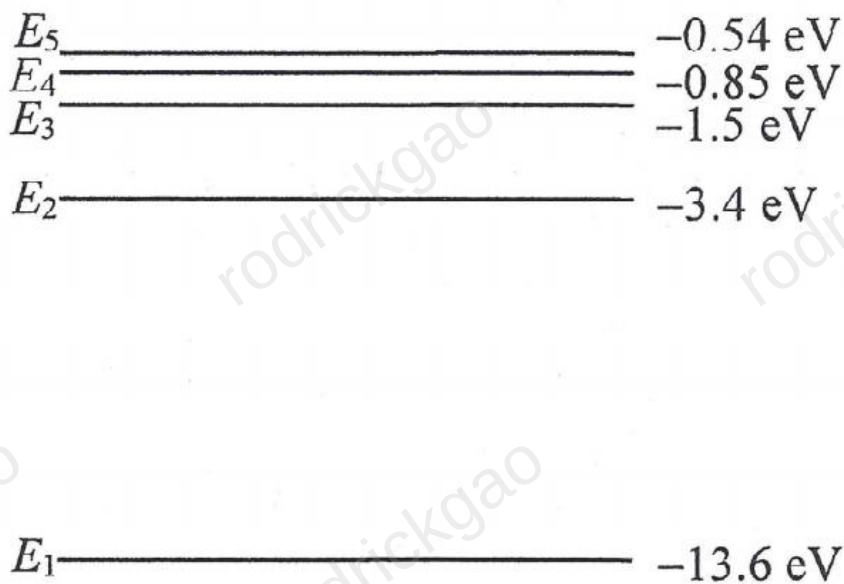
- A slope unchanged; intercept larger
- B slope unchanged; intercept smaller**
- C slope smaller; intercept larger
- D slope larger; intercept smaller

Official answer: B

$K_{\max} = hc(1/\lambda) - \phi$, so the slope hc is unchanged by intensity or metal. A smaller work function gives a smaller horizontal intercept $\phi/(hc)$. Answer: B.

2.5

Hydrogen energy levels E_1 to E_5 are shown. The transition $E_4 \rightarrow E_2$ emits blue light. Which transition emits red light?



A $E_5 \rightarrow E_2$

B $E_4 \rightarrow E_3$

C $E_3 \rightarrow E_2$

D $E_2 \rightarrow E_1$

Official answer: C

Red light has lower photon energy than blue light. Among the listed Balmer transitions, $E_3 \rightarrow E_2$ has a smaller energy difference than $E_4 \rightarrow E_2$ and therefore emits red light. Answer: C.

2.6

For an electron accelerated through voltage V , $\lambda = h/\sqrt{(2meV)}$. If λ is in nm and V in kV, λ is approximately

A $0.04/\sqrt{V}$

B $0.12/\sqrt{V}$

C $0.4/\sqrt{V}$

D $1.2/\sqrt{V}$

Official answer: A

Substituting the constants in $\lambda = h/\sqrt{(2meV)}$ and expressing V in kV and λ in nm gives $\lambda \approx 0.04/\sqrt{V}$. Answer: A.

2.7

Which statements about an optical microscope and a transmission electron microscope (TEM) are correct?

- (1) The higher resolving power of TEM is enabled by the much shorter wavelength of its electron beam.
- (2) Current-carrying coils in a TEM provide magnetic fields for converging the electron beam, like a lens converging light.
- (3) The angular resolution of both microscopes is limited by Rayleigh's criterion.

- A (1) and (2) only
- B (1) and (3) only
- C (2) and (3) only
- D (1), (2) and (3)**

Official answer: D

A TEM gains resolving power from the electron's short wavelength; magnetic fields from coils act as lenses; and diffraction sets the angular-resolution limit represented by Rayleigh's criterion. All three statements are correct. Answer: D.

2.8

A nano material

- (1) has a higher volume-to-surface-area ratio than the same substance in bulk form.
- (2) has at least one dimension less than 1 nm.
- (3) is chemically more reactive than the same substance in bulk form.

- A (1) only
- B (3) only**
- C (1) and (2) only
- D (2) and (3) only

Official answer: B

A nanomaterial has a larger surface-area-to-volume ratio, not the reverse; its nanoscale dimension is typically about 1–100 nm, not necessarily below 1 nm. Its high surface exposure can make it more reactive. Only statement (3) is correct. Answer: B.

Structured question

- (a) Rutherford's planetary model failed to account for the stability of atoms. Why? (1 mark)

- (b) The emission spectrum of hydrogen has four visible lines A to D. These transitions end at the first excited state ($n = 2$). The wavelengths are given by $\lambda = 364.6[n^2/(n^2 - 2^2)]$ nm, where $n = 3, 4, 5, \dots$

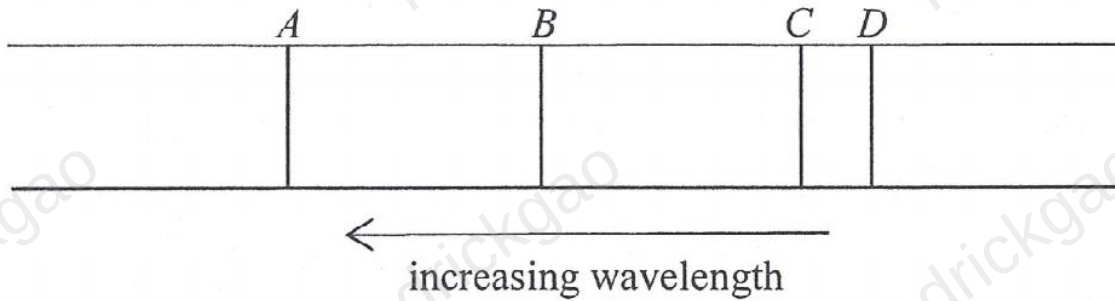
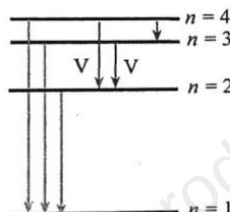


Figure 2.1

- (b)(i) Which line (A, B, C or D) comes from the transition $n = 5$ to $n = 2$? (1 mark)
- (b)(ii) Find its wavelength and state its colour. (2 marks)
- (b)(iii) A photon of wavelength shorter than 364.6 nm strikes a hydrogen atom in the first excited state. State what happens to the photon, the atom and its orbital electron. (3 marks)
- (b)(iv) Hydrogen atoms are initially in the third excited state ($n = 4$). On an energy-level diagram show ALL possible emission transitions, and mark V beside those giving visible lines. (3 marks)

Official marking scheme

2. (a)	In the model the orbital electron gradually loses its energy (via radiation) and spirals inwards until it finally crashes into the nucleus, i.e. the atom collapses.	1A	
			1
(b) (i)	Line C (from $n = 5$ to $n = 2$)	1A	1
(ii)	$\lambda = 364.6 \left(\frac{5^2}{5^2 - 2^2} \right)$ $= 434.047619 \text{ (nm)} \approx 434 \text{ (nm)}$ Colour – violet / blue / indigo	1A	
		1A	2
(iii)	The incident photon is absorbed. The hydrogen atom is ionized / becomes a hydrogen ion. The orbital electron becomes a free electron, (i.e. liberated with some kinetic energy).	1A	
		1A	3
(iv)	 (Total number of spectral lines = 6)	2A	
		1A	
			3
			Two visible lines correctly marked.

(a) An accelerating orbital electron would radiate energy, spiral inward and eventually fall into the nucleus, so Rutherford's model cannot explain atomic stability.

(b)(i) The $n = 5 \rightarrow 2$ transition is line C.

(ii) $\lambda = 364.6 [5^2 / (5^2 - 2^2)] = 434 \text{ nm}$, violet/blue.

(iii) A photon with $\lambda < 364.6 \text{ nm}$ has enough energy to be absorbed and ionize the atom; the electron is liberated with kinetic energy.

(iv) From $n = 4$, show all six downward transitions: $4 \rightarrow 3$, $4 \rightarrow 2$, $4 \rightarrow 1$, $3 \rightarrow 2$, $3 \rightarrow 1$ and $2 \rightarrow 1$. Mark the visible Balmer transitions $4 \rightarrow 2$ and $3 \rightarrow 2$ with V.

Section C: Energy and Use of Energy

Multiple-choice questions

3.1

Green light has a maximum luminous efficacy of 683 lm W^{-1} . A filament lamp emits 1 W of white light producing half this luminous flux, and only 3% of its input energy becomes light. Estimate its luminous efficacy.

- A 40 lm W^{-1}
- B 20 lm W^{-1}
- C 10 lm W^{-1}
- D 5 lm W^{-1}

Official answer: C

The lamp produces about $683/2 = 341.5 \text{ lm}$ from 1 W of light. At 3% efficiency, 1 W of light requires 33.3 W input, so efficacy = $341.5/33.3 \approx 10 \text{ lm W}^{-1}$. Answer: C.

3.2

A wind turbine generator can extract energy from moving air. However, the kinetic energy of wind cannot be fully transformed into electrical energy because

- (1) wind velocity cannot be zero after passing through the turbine.
- (2) there is loss in transformation of energy in the generator.
- (3) wind may not always blow in the direction normal to the turbine.

- A (1) and (2) only
- B (1) and (3) only
- C (2) and (3) only
- D (1), (2) and (3)

Official answer: D

Wind must retain some speed after the turbine, the generator has conversion losses, and non-normal wind reduces the extractable component. All three prevent complete conversion. Answer: D.

3.3

The atmosphere absorbs 26.8% of the Sun's radiation. A 5.0 m^2 solar panel has efficiency 15%. The solar constant is 1366 W m^{-2} . Estimate its electrical output.

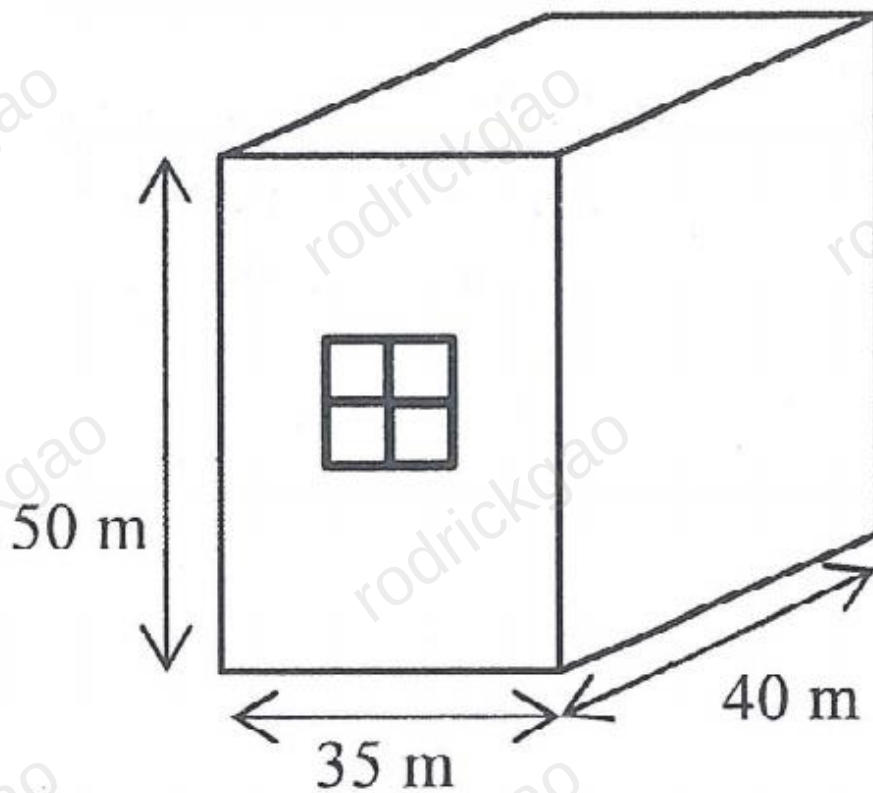
- A 275 W
- B 750 W**
- C 1560 W
- D 4250 W

Official answer: B

$$P = 1366(5.0)(1 - 0.268)(0.15) = 7.50 \times 10^2 \text{ W. Answer: B.}$$

3.4

A rectangular building is $35\text{ m} \times 40\text{ m} \times 50\text{ m}$. Its overall thermal transfer value must not exceed 24 W m^{-2} . Windows are $2\text{ m} \times 3\text{ m}$. For $\Delta T = 10\text{ K}$, $U(\text{concrete}) = 2.0\text{ W m}^{-2}\text{ K}^{-1}$ and $U(\text{glass}) = 5.7\text{ W m}^{-2}\text{ K}^{-1}$. Find the allowable number of windows.



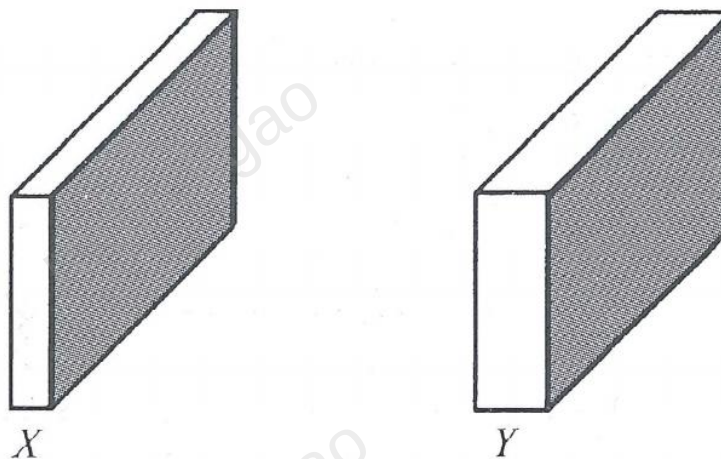
- A 960
- B 598
- C 160
- D 120

Official answer: C

Using $OTTV = [U_{cA_c}\Delta T + U_{gA_g}\Delta T]/A_{\text{wall}}$ with each window area 6 m^2 and the dimensions shown, imposing $OTTV \leq 24\text{ W m}^{-2}$ gives a maximum of 160 windows. Answer: C.

3.5

Walls X and Y have the same material and area, but Y is thicker. They have the same temperature difference. Which quantities are the same?



(1) thermal conductivity

- (2) U-value
- (3) rate of heat transfer

- A (1) only
- B (3) only
- C (1) and (2) only
- D (2) and (3) only

Official answer: A

Thermal conductivity is a material property, so it is the same. Since $U = k/d$, the thicker wall has smaller U and therefore a smaller conduction rate $UA\Delta T$. Only statement (1) is correct. Answer: A.

3.6

An air-conditioner uses 420 kWh in 1200 h and has cooling capacity 2.54 kW. Find its coefficient of performance.



energy consumption of 420 kW h
for 1200 hours operation per year

cooling capacity
2.54 kW

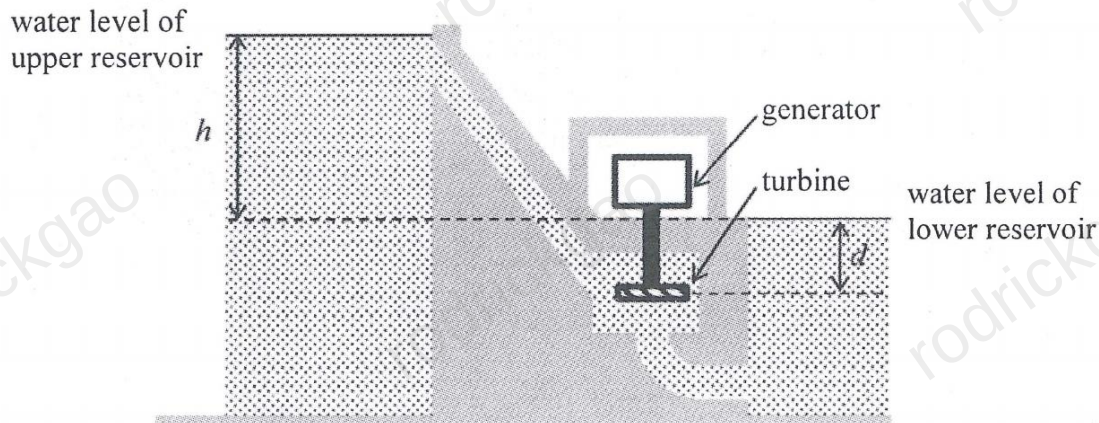
- A 1.12
- B 1.38
- C 7.26**
- D 8.89

Official answer: C

Average electrical input power = $420 \text{ kW h} / 1200 \text{ h} = 0.350 \text{ kW}$. Therefore $\text{COP} = 2.54 / 0.350 = 7.26$. Answer: C.

3.7

Which factors below can affect the maximum power output of the hydroelectric power plant?



(1) The height difference between the water levels in the upper and lower reservoirs, h .

(2) The distance between the turbine and the water level of the lower reservoir, d .

(3) The rate of water flowing through the turbine.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

Official answer: B

Hydroelectric power is approximately ρQgh , so it depends on the head h and the water-flow rate, not on the turbine's distance d above the lower reservoir. Thus (1) and (3) are correct. Answer: B.

3.8

The mass defect in one fission of uranium-235 is 0.1855 u. Which expression gives the energy from 1 kg of uranium-235? Molar mass = 235 g mol⁻¹; 1 u corresponds to 1.49×10^{-10} J.

A $(1000/235)(6.02 \times 10^{23})(0.1855)(1.49 \times 10^{-10})$

B $(1/235)(6.02 \times 10^{23})(0.1855)(1.49 \times 10^{-10})$

C $(235/1000)(6.02 \times 10^{23})(0.1855)(1.49 \times 10^{-10})$

D $(1000/235)(6.02 \times 10^{23})(1.49 \times 10^{-10})$

Official answer: A

The number of U-235 nuclei in 1 kg is $(1000/235)N_A$. Multiplying this by the mass defect 0.1855 u and 1.49×10^{-10} J per u gives expression A. Answer: A.

Structured question

Two electric vehicles have the following data: A—battery capacity 95 kWh, maximum range 326 km, mass 2500 kg; B—battery capacity 66 kWh, maximum range 414 km, mass 1620 kg.

electric vehicle	battery capacity / kW h	maximum driving range / km	mass / kg
<i>A</i>	95	326	2500
<i>B</i>	66	414	1620

- (a) Vehicle A has a larger battery capacity but a shorter range. Give one reason and explain. (1 mark)
- (b)(i) Vehicle A is charged from a 220 V supply. Find the minimum current needed for a full charge in 12 h. (2 marks)
- (b)(ii) Explain why the practical charging current must be larger. (1 mark)

Performance data: A—0 to 100 km h⁻¹ in 5.5 s, peak motor power 300 kW; B—0 to 100 km h⁻¹ in 6.5 s, peak motor power 150 kW.

electric vehicle	time required to accelerate from 0 to 100 km h ⁻¹ / s	peak power / kW
<i>A</i>	5.5	300
<i>B</i>	6.5	150

- (c)(i) Assuming the motor of A delivers peak power throughout acceleration, estimate its energy efficiency. (2 marks)
- (c)(ii) In the maximum-range test, B travels at an average speed of 70 km h⁻¹. Estimate its average battery output power. (2 marks)

- (d) Compare the effectiveness of regenerative braking in: stop-go motion at a few km h^{-1} ; city driving with smooth traffic-light flow; and highway driving. (2 marks)

Mode 1	driving at a few km per hour in often stop-and-go traffic conditions
Mode 2	driving in a city with smooth traffic regulated by traffic lights
Mode 3	driving on a highway

Official marking scheme

3. (a)	resistance / friction.		1A	
	Or			
	The mass of A is larger, more energy is dissipated in accelerating and/or decelerating the vehicle.			
	(Accept reasonable factors relating to conversion efficiency)			
			1	
(b) (i)	$95 \times 10^3 \text{ W h} = 220 \text{ V} \times I \times 12 \text{ h}$ $I = 35.984848 \text{ A} \approx 36.0 \text{ A}$	1M 1A		Accept 35.9 A – 36.1 A
			2	
(ii)	The charging efficiency is not 100% / energy is lost in the charging process (as heat / thermal energy).	1A		
			1	
(c) (i)	Power out = $\frac{\frac{1}{2}mv^2}{t} = \frac{\frac{1}{2} \times 2500 \times (\frac{100}{3.6})^2}{5.5} \approx \frac{9.64506 \times 10^5}{5.5}$ $= 1.753648 \times 10^5 \text{ W} \approx 175 \text{ kW}$ Efficiency = $\frac{175}{300} \times 100\%$ $= 58.454920\% \approx 58.5\%$	1M 1A		Accept 58% – 59%
			2	
(ii)	Total time taken for maximum driving range test $= \frac{414}{70} = 5.914286 \text{ h} \approx 5.91 \text{ h}$ Power output = $\frac{66}{5.91}$ $= 11.159420 \text{ kW} \approx 11.2 \text{ kW}$	1M 1A		Accept 11.0 kW – 11.5 kW
			2	
(d)	Mode 2 (driving in a city with smooth traffic regulated by traffic lights) as (regenerative) braking can utilize the (relatively large) kinetic energy of the vehicle, say, when stopping at traffic lights.	1A+1A		
	Or			
	Mode 1 (driving at a few km per hour in often stop-and-go traffic conditions) as (regenerative) braking needs to be applied often.	0A+1A		
			2	

(a) Vehicle A is much heavier, so more energy is required against resistance and for repeated acceleration; this can shorten its range despite the larger battery.

(b)(i) $95 \times 10^3 \text{ W h} = (220 \text{ V})I(12 \text{ h})$, so $I = 36.0 \text{ A}$.

(ii) Charging is not 100% efficient, so the input current must supply the energy losses.

(c)(i) At $100 \text{ km h}^{-1} = 27.8 \text{ m s}^{-1}$, useful average power = $[\frac{1}{2}(2500)(27.8^2)]/5.5 = 175 \text{ kW}$. Efficiency = $175/300 = 58.5\%$.

(ii) Test time = $414/70 = 5.91 \text{ h}$, so average battery output = $66/5.91 = 11.2 \text{ kW}$.

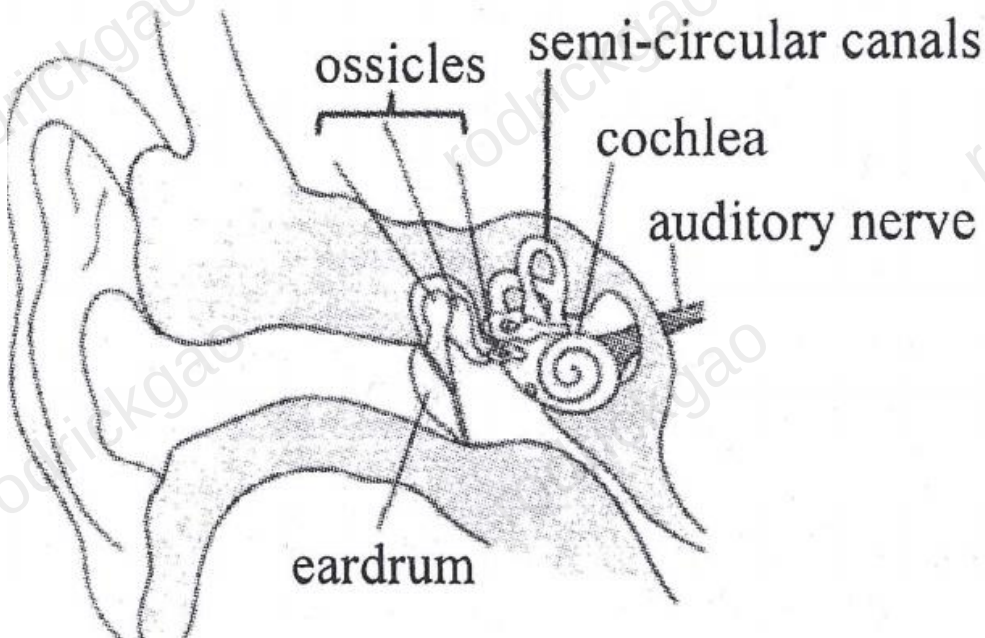
(d) Mode 2 is most effective: braking from normal city speeds at traffic lights recovers substantial kinetic energy. Mode 1 has little kinetic energy at a few km h^{-1} , while mode 3 requires little braking.

Section D: Medical Physics

Multiple-choice questions

4.1

Which part of the ear distinguishes different sound frequencies?



- A eardrum
- B semicircular canals
- C ossicles
- D cochlea

Official answer: D

Frequency discrimination occurs in the cochlea, where different positions respond most strongly to different frequencies. Answer: D.

4.2

Each optical fibre in an endoscope consists of a core enclosed by a cladding. Which descriptions about an optical fibre are correct?

- (1) The refractive index of the cladding is smaller than that of the core.
- (2) The core-cladding boundary gives a smaller critical angle compared to a core-air boundary.
- (3) Without cladding, some of the light rays would pass between optical fibres at points of contact.

- A (1) and (2) only
- B (1) and (3) only**
- C (2) and (3) only
- D (1), (2) and (3)

Official answer: B

The core index exceeds the cladding index, and cladding prevents light leaking between touching fibres.

Because $n_{\text{cladding}} > n_{\text{air}}$, the core-cladding critical angle is larger, not smaller, than the core-air value. Thus (1) and (3) are correct. Answer: B.

4.3

Which statements comparing A-scan and B-scan ultrasound are correct?

- (1) B-scan can locate tumours.
- (2) B-scan can display movement in real time.
- (3) B-scan has higher spatial resolution than A-scan.

- A (1) and (2) only**
- B (1) and (3) only
- C (2) and (3) only
- D (1), (2) and (3)

Official answer: A

B-scans form two-dimensional images useful for locating tumours and can display organ motion in real time. A

B-scan is not inherently of higher spatial resolution than an A-scan. Thus (1) and (2) are correct. Answer: A.

4.4

For locating internal bleeding in the brain, which are valid reasons for NOT using the stated method?

- (1) X-ray radiography: insufficient resolution.
- (2) Ultrasound scanning: ultrasound cannot penetrate the skull.
- (3) Endoscopy: there is no cavity in the brain for inserting an endoscope.

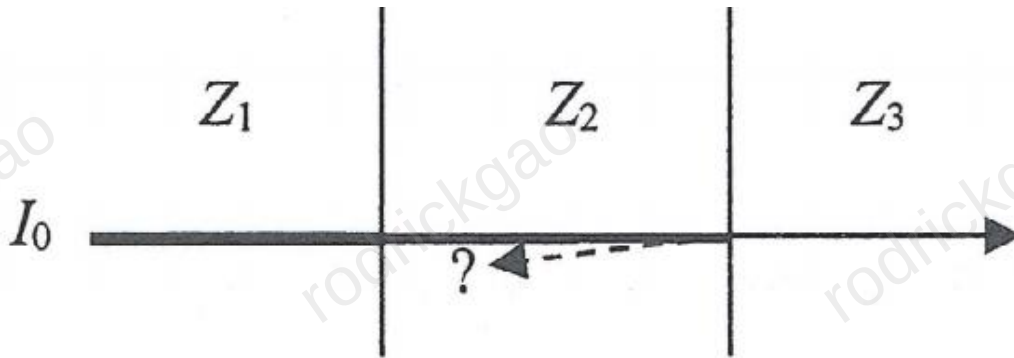
- A (1) and (2) only
 B (1) and (3) only
 C (2) and (3) only
 D (1), (2) and (3)

Official answer: C

Ultrasound is blocked strongly by the skull, and there is no brain cavity suitable for inserting an endoscope. The stated 'insufficient resolution' is not the correct reason by itself for rejecting radiography. Thus (2) and (3) are correct. Answer: C.

4.5

Ultrasound of intensity I_0 passes through media of acoustic impedances Z_1 , Z_2 and Z_3 . Neglect attenuation and absorption. What is the intensity reflected from the Z_2 – Z_3 interface?



- A $[1 - (Z_2 - Z_1)^2 / (Z_2 + Z_1)^2] \cdot (Z_3 - Z_2)^2 / (Z_3 + Z_2)^2 \cdot I_0$
 B $(Z_2 - Z_1)^2 / (Z_2 + Z_1)^2 \cdot (Z_3 - Z_2)^2 / (Z_3 + Z_2)^2 \cdot I_0$
 C $(Z_2 - Z_1)^2 / (Z_2 + Z_1)^2 \cdot [1 - (Z_3 - Z_2)^2 / (Z_3 + Z_2)^2] \cdot I_0$
 D $[1 - (Z_2 - Z_1)^2 / (Z_2 + Z_1)^2] \cdot [1 - (Z_3 - Z_2)^2 / (Z_3 + Z_2)^2] \cdot I_0$

Official answer: A

The intensity reaching the second interface is $I_0[1 - ((Z_2 - Z_1)/(Z_2 + Z_1))^2]$. Multiplying by the reflection coefficient at the Z_2 – Z_3 boundary, $((Z_3 - Z_2)/(Z_3 + Z_2))^2$, gives option A. Answer: A.

4.6

The intensity of an X-ray beam is decreased by 25% after passing through a 0.010 m metal plate. Find the corresponding half-value thickness.

- A 0.005 m
- B 0.020 m
- C 0.024 m**
- D 0.042 m

Official answer: C

Using $I = I_0 e^{-\mu x}$, $\mu = -\ln(0.75)/0.010$. Hence $x_{1/2} = \ln 2 / \mu = 0.0241 \text{ m} \approx 0.024 \text{ m}$. Answer: C.

4.7

Radionuclide imaging uses only γ radiations as

- (1) γ can be deflected by a magnetic field to be incident on the patient at any angle.
- (2) γ has low ionizing power and causes less harm to cells.
- (3) γ has high penetrating power and is detectable outside the body.

- A (1) only
- B (3) only
- C (1) and (2) only
- D (2) and (3) only**

Official answer: D

Gamma rays are not deflected by a magnetic field. Their relatively low ionizing power limits tissue damage and their high penetrating power lets them leave the body for detection. Thus (2) and (3) are correct. Answer: D.

4.8

Radiation weighting factors are 20 for α particles and 1 for β particles, γ rays and X-rays. Why is the α value larger?

- A α particles have low penetration.
- B α particles are strongly ionizing.**
- C α particles have relatively large mass because they are helium nuclei.
- D α radiation has a particle nature.

Official answer: B

Alpha particles are densely ionizing and cause much greater biological damage per unit absorbed energy, so their radiation weighting factor is larger. Answer: B.

Structured question

- (a) Roger wears bifocal spectacles. The upper half has power -1.0 D and the lower half $+2.0\text{ D}$. With the spectacles, his near point is 25 cm and his far point is infinity. The lenses are close to his eyes.

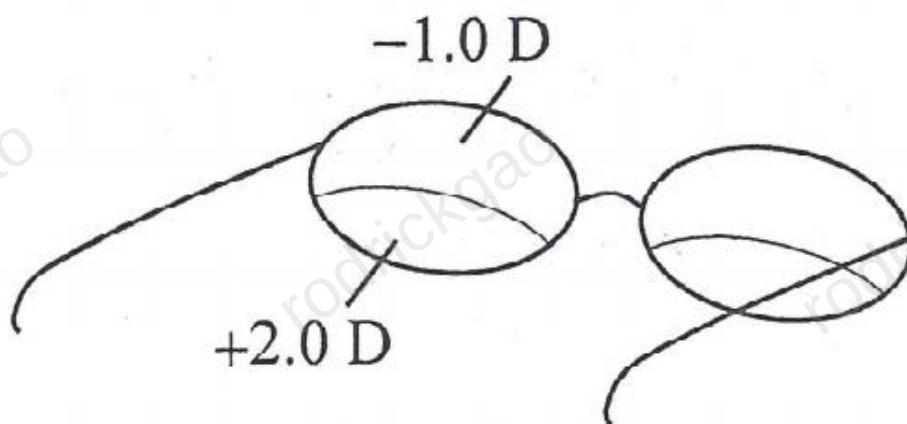


Figure 4.1

- (a)(i) State which half is used for viewing distant objects and determine Roger's unaided far point. (2 marks)
- (a)(ii) An object O is 25 cm in front of the lens. (1) Complete the ray diagram to show where the image forms on the retina, and indicate the unaided near point N . (2 marks) (2) Find the distance of N from the eye. (2 marks)

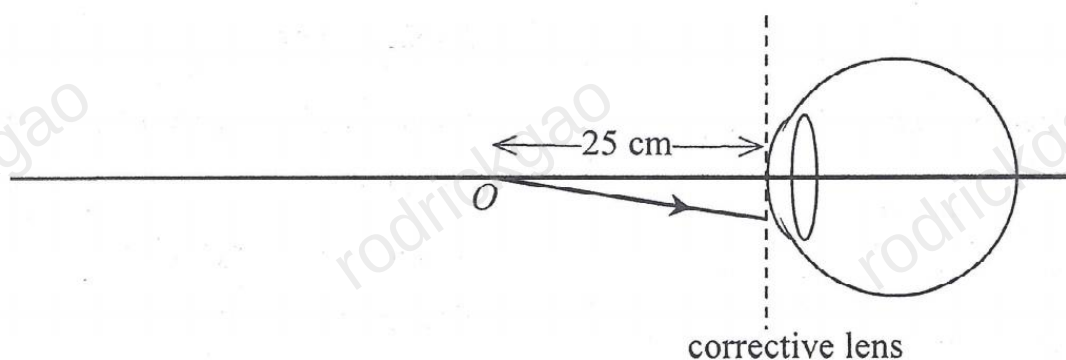


Figure 4.2

- (b) An ultrasound A-scan of an eye is shown. The labelled echoes correspond to interfaces A to D.

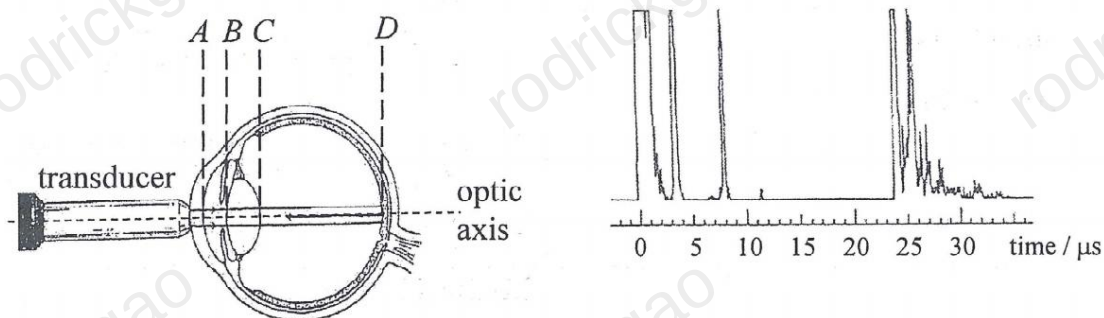
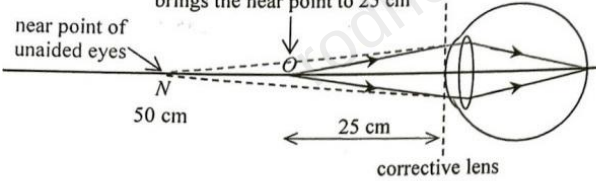


Figure 4.3

- (b)(i) Determine the thickness of the eye lens. Take the speed of ultrasound in the lens as 1520 m s^{-1} . (2 marks)
- (b)(ii) State whether 3 MHz or 15 MHz ultrasound is preferred and explain. (1 mark)
- (b)(iii) State one medical application of ultrasound other than imaging. (1 mark)

Official marking scheme

$\text{Power} = \frac{1}{f} = -1.0 \text{ D} \Rightarrow -1.0 = \frac{1}{\infty} + \frac{1}{-d}$ $\therefore \text{Far point distance of unaided eyes } d = 1.0 \text{ m} = 100 \text{ cm}$		1A/1M	$P_{\text{eye}} = \frac{1}{d} + \frac{1}{D_{\text{eyeball}}} \quad (1)$ $P_{\text{eye}} - 1.0 = \frac{1}{\infty} + \frac{1}{D_{\text{eyeball}}} \quad (2)$	(1 M)
		2		
(ii) (1) The converging / corrective lens brings the near point to 25 cm 		2A		
		2		
(2) With the corrective lens (convex lens), object at $O \Rightarrow u = 25 \text{ cm} = 0.25 \text{ m}$ virtual image at $N \Rightarrow v = -d$ (where d is the near point distance of unaided eyes) $P = \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ $\Rightarrow 2 = \frac{1}{0.25} + \frac{1}{-d}$ $\Rightarrow d = 0.5 \text{ m} = 50 \text{ cm}$		1M	Alternative method: $P_{\text{eye}} = \frac{1}{d} + \frac{1}{D_{\text{eyeball}}} \quad (1)$ $P_{\text{eye}} + 2 = \frac{1}{0.25} + \frac{1}{D_{\text{eyeball}}} \quad (2)$	(1M)
		1A		
		2		
(b) (i) $\Delta t = t_2 - t_1 = 8 - 3 = 5 \mu\text{s}$ $d = \frac{1}{2} c \Delta t$ OR $d = ct'$ where $t' = \frac{\Delta t}{2}$ $= \frac{1}{2} (1520)(5 \times 10^{-6}) = 3.8 \times 10^{-3} \text{ m} = 3.8 \text{ mm}$		1M		Accept $4 \mu\text{s}$ to $5 \mu\text{s}$
		1A		Accept 3.04 mm ($\Delta t = 4 \mu\text{s}$) to 3.8 mm
		2		
(ii) The ultrasound pulses only need to detect structures near surface, absorption and attenuation are small. 15 MHz ($\lambda \sim 0.023 \text{ mm}$) should be chosen as its wavelength is shorter than 3 MHz ($\lambda \sim 0.11 \text{ mm}$), (spatial) resolution is higher.		1A		
		1		
(iii) Any ONE of the applications: - break up stones in kidney, gall bladder etc. - break down crystalline eye lens in cataract surgery - to remove calculus from surface of teeth in dentistry - high intensity ultrasound to destroy tumours (surgery) - speed up bone healing after a fracture. - therapeutic uses: reduce pain, increase circulation and increase mobility of soft tissues, reduce inflammation and healing of injuries and wounds.		1A		
		1		

(a)(i) The upper -1.0 D half corrects distant vision. Its focal length is -1.0 m , so Roger's unaided far point is 1.0 m .

(ii) (1) The $+2.0 \text{ D}$ lower half forms a virtual image at Roger's unaided near point N; complete the ray so the eye

focuses it on the retina and mark N 50 cm in front of the eye. (2) With $u = 0.25$ m and $v = -d$, $1/f = 1/u + 1/v$ gives $2.0 = 4.0 - 1/d$, so $d = 0.50$ m.

(b)(i) The lens echoes are separated by about $5\text{ }\mu\text{s}$, so thickness $= \frac{1}{2}c\Delta t = \frac{1}{2}(1520)(5 \times 10^{-6}) = 3.8 \times 10^{-3}\text{ m} = 3.8\text{ mm}$.

(ii) Use 15 MHz: attenuation is acceptable over the shallow eye and its shorter wavelength gives better spatial resolution.

(iii) One valid use is breaking kidney stones, cataract surgery, dental scaling, tumour ablation, or therapeutic promotion of healing.